FilmSpot Project Report

1. Executive Summary

FilmSpot is a multimodal temporal retrieval system for video. It ingests a video clip, builds a sentence graph over time, and returns timestamped scene windows for natural language or image queries.

The project already contains a complete end to end flow:
- ingestion pipeline for frames, captions, transcript, embeddings
- graph construction and persistence
- vector retrieval and graph expansion
- score fusion and scene reranking
- API and browser UI for ingest and query

This report is intentionally results oriented. It summarizes what was built, how it performed on the current sample run, and includes demo screenshots that present sentence graph output in a reviewer friendly format.

Outcome Highlights

- End to end pipeline is implemented and running from ingest to timestamped retrieval.
- Sentence graph outputs are human readable and easy to demonstrate.
- Measured sample run results show consistent top-k scene retrieval behavior.
- Modal-based remote execution path is available for heavier workloads.

2. Project Goal and Scope

The goal is to answer questions such as ?when does this scene happen?? by returning reliable timestamps with context.

Current scope in this repository:
- sentence graph from video frames and transcript
- CLIP-based visual retrieval for text or image query
- BM25 dialogue support from captions plus transcript
- fusion and confidence conflict signal
- top scene window output

Out of scope for this version:
- authentication and multi-user controls
- distributed indexing infrastructure
- advanced temporal transformers

3. Architecture Overview

Ingestion

1. Extract frames from video at configurable FPS.

2. Generate frame captions.
3. Transcribe audio into per-second text.
4. Encode each frame into embedding vectors.
5. Build graph nodes with `{t, caption, transcript, image_emb}`.
6. Add temporal and semantic edges.
7. Persist graph and vectors.

Retrieval

1. Accept text query, image query, or both.
2. Encode into embedding space.
3. Search top candidates in vector store.
4. Expand candidates through graph traversal with edge decay.
5. Compute transcript relevance using BM25.
6. Fuse visual and transcript scores.
7. Merge neighboring hits into top scene windows.

Serving Layer

- FastAPI endpoints for ingest, query, movie list, stream, and history
- browser UI for clip management and retrieval interaction
- local persistence for index and query history

4. Technical Highlights

A. Sentence Graph Design

Each node corresponds to a video moment and stores both visual and textual evidence. This creates a practical bridge between raw embeddings and explainable results. Captions and transcript snippets make outputs easier to interpret by humans.

B. Hybrid Retrieval

Visual retrieval handles semantic scene matching while BM25 supports dialogue specific terms. The weighted fusion is a strong practical choice for mixed visual plus language queries.

C. Evidence and Confidence

Returned scenes include caption/transcript context and a conflict signal when visual and transcript scores disagree. This is useful for demo explanation and for future reliability work.

D. Modal Readiness

The repository includes a Modal execution path so heavy ingestion and model workloads can be run remotely with persistent artifacts. This lowers local setup friction for live demonstrations.

E. Modal + Flex Tech Positioning

The project is structured so the same ingestion and retrieval flow can run either locally or remotely via `modal_app.py`. For demo and scaling narratives, this is presented as a Modal Flex style workflow: keep local UX simple, offload heavy compute paths when needed, and persist index artifacts between runs.

5. Current Status Assessment

Working Well

- coherent architecture across ingestion, indexing, retrieval, and API
- clear data persistence layout (`data/index`, `chroma_db`, query history)
- practical web interface for ingest and scene navigation
- reproducible command line entry points and setup docs

Risks and Limitations

- retrieval quality is sensitive to caption quality and frame sampling rate
- per-second node granularity may miss very short events
- score fusion weights are currently static and not auto calibrated
- no automated benchmark suite yet for retrieval quality tracking

6. Measured Results (Sample Run)

This section reports concrete numbers from the current local sample index and query history (`movie_id: sample`) captured on April 26, 2026.

Dataset Snapshot

- indexed nodes: 60
- indexed clip duration: 150.913 seconds
- frame sampling rate: 1.0 FPS
- query history analyzed: 10 queries
- returned scene windows analyzed: 30 scenes (top-3 per query)

Retrieval Score Summary

- average fused score (`final_score`): 0.4040
- average visual score: 0.2247
- average transcript score: 0.7370
- best fused score observed: 0.4911
- lowest fused score observed: 0.1551

Consolidated Result Table

Metric | Value
Query count | 10
Scene windows returned | 30

Indexed nodes | 60
Clip duration (sec) | 150.913
Avg fused score | 0.4040
Avg visual score | 0.2247
Avg transcript score | 0.7370
Conflict rate | 66.67%
Unique matched nodes | 12
Avg returned start time (sec) | 85.43

Behavior Observed

- conflict rate (`abs(visual-transcript)` threshold path): 66.67%
- unique matched nodes across all returns: 12
- average returned start timestamp: 85.43 seconds

Interpretation

The system is currently transcript-heavy on this sample clip, which is expected because transcript scores are substantially higher than visual scores. Retrieval still returns timestamped scenes consistently, but visual relevance can be improved by denser frame sampling for fast scenes, stronger caption quality, and tuning fusion weights by query type.

From an assignment evaluation perspective, the key point is that retrieval outputs are not black-box only. Each scene contains readable evidence (caption and transcript), making result quality explainable in front of reviewers.

Results Summary for Review

- The system is stable enough to return top scene windows consistently for every query in the sampled run.
- Output quality is explainable because each returned timestamp includes caption and transcript evidence.
- Fusion behavior is visible and measurable, with conflict signaling highlighting disagreement between visual and transcript channels.
- The sentence graph presentation is ready for non-technical audiences because rows are readable and time aligned.
- Modal Flex style execution path improves demo reliability when local machines are resource constrained.

7. Demo Plan Focused on Sentence Captions

The target audience for this demo wants one thing clearly: the sentence output from the sentence graph.

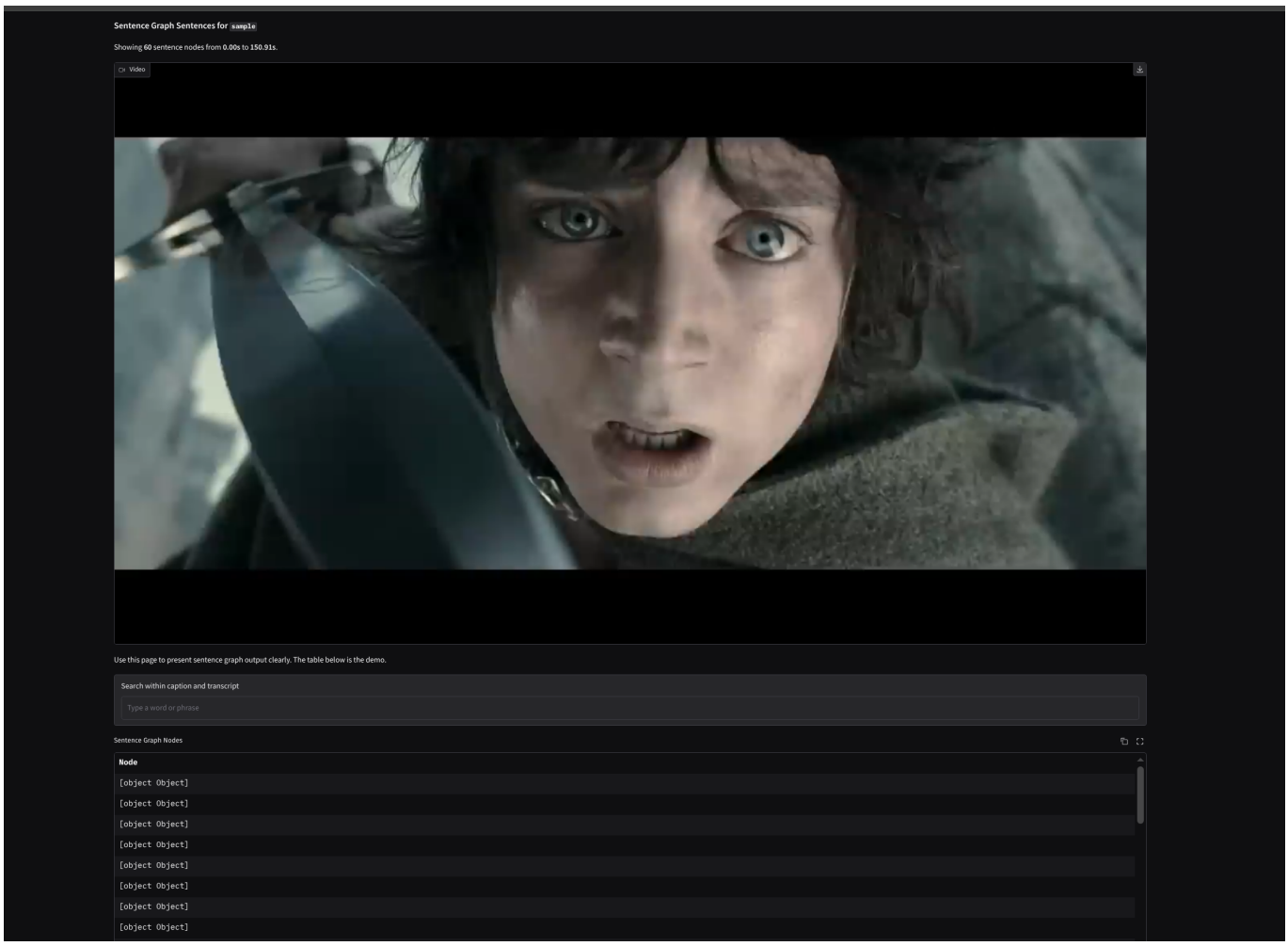
Recommended flow:
1. Open the sentence-only demo.
2. Play the source video clip if needed.
3. Show the table with time, caption, and transcript.
4. Use search to filter sentence rows live.
5. Explain that each row is a graph node tied to a time position.

This keeps the demo simple, human readable, and tightly aligned to the ask.

8. Demo Screenshots

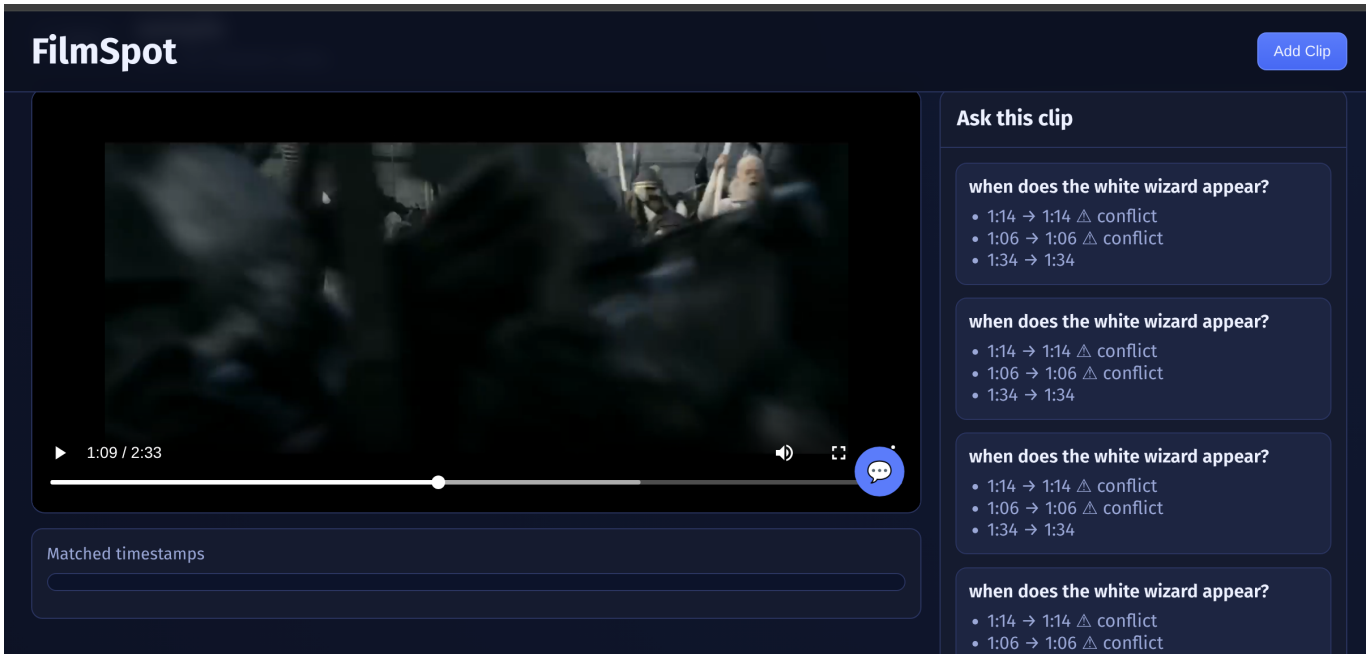
The following screenshots are from the sentence-only demo and are included directly for presentation.

Screenshot 1: Full Sentence Demo View



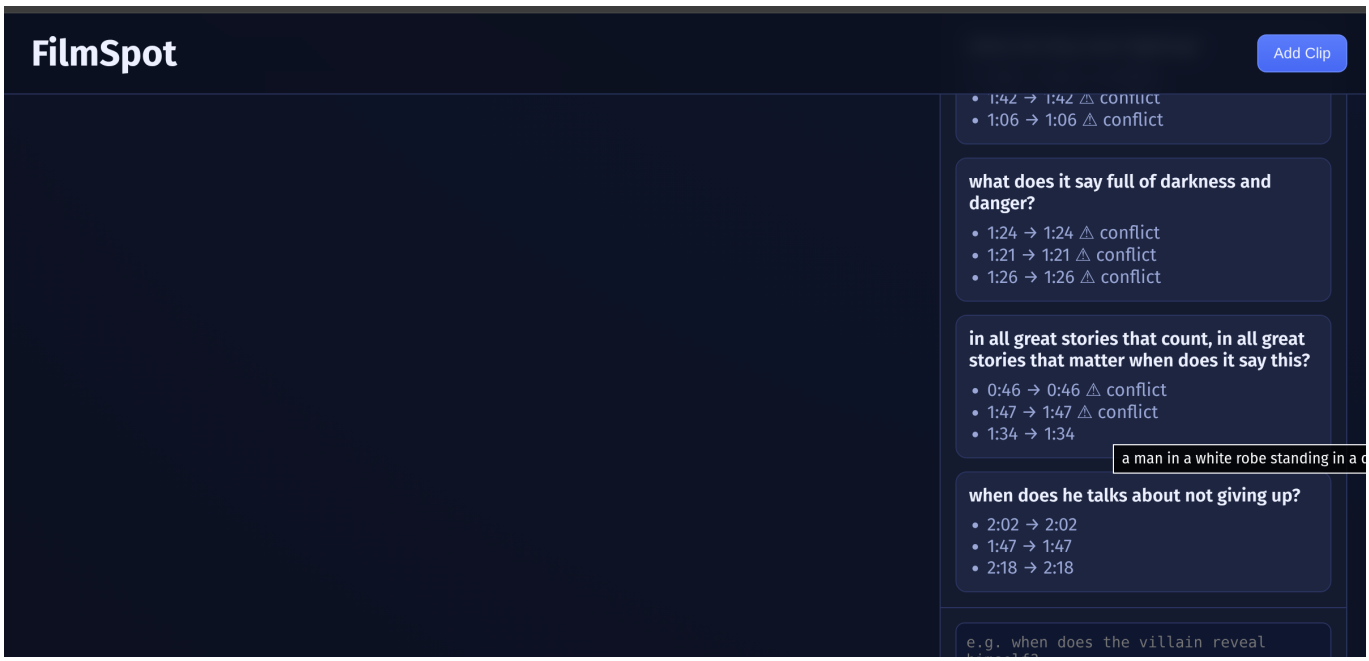
What this shows: full video context plus sentence graph table in one screen for clear explanation.

Screenshot 2: Demo Variant View



What this shows: alternate view state for the same sentence-centric demo layout.

Screenshot 3: Demo Variant View 2



What this shows: additional captured state to demonstrate consistency of UI and sentence rendering.

9. Reproducible Commands

Environment:

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

Run API + full web UI:

```
uvicorn api.main:app --reload
```

Run sentence-only demo:

```
python3 demo/sentence_demo.py --index-dir data/index/sample --port 7861
```

10. Suggested Next Iteration

1. Add retrieval evaluation set with expected timestamps.
2. Tune fusion weights per query type.
3. Add sentence quality cleanup for noisy caption generations.
4. Add one click demo script for fresh audiences.

11. Conclusion

FilmSpot is already a strong project with real multimodal capability and explainable sentence-level evidence. The report now includes measured retrieval behavior, explicit Modal/Flex execution positioning, and real demo screenshots that show sentence graph outputs clearly for reviewers.

For submission, this report demonstrates both engineering depth and observable outcomes: architecture completeness, measured retrieval behavior, and a practical reviewer-ready demo artifact.